

# Progress Page

## Student Information

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- <https://github.com/lohnerj/ase-392ResearchProject>  (<https://github.com/lohnerj/ase-392ResearchProject>)

## Planning the Summer Project

### Summary of the Project

Use machine learning in order to predict if the price of various items will raise or drop in the future.

#### Goals

1. Develop the architecture/design for the database and front-end interface.
2. Enhance/change database structure and optimize for more data/machine learning applications.
3. Implement the front-end interface for the machine learning module.
4. Expand the API suite to support the upgraded database functionalities.
5. Integrate new APIs into the front-end system.
6. Evaluate various machine learning algorithms for optimal performance.
7. Integrate the selected machine learning algorithm into the production codebase.
8. Assemble all components into a cohesive system.
9. Refactor code for efficiency and maintainability.
10. Complete and refine the project documentation.
11. Conduct backend testing.
12. Conduct front-end testing.

To create an easily accessible user interface that allows for users to easily see and use market information.

#### Why?

Create a useful tool for people to manage the complexity of the in-game market, while also providing useful analytics.

## Features

### User Story Requirements:



1.0: As a developer, I want to develop the architecture for the database and front-end interface so that we have a robust foundation for the application.

1.1: As a data engineer, I want to enhance the database structure and optimize it to handle more data and support machine learning applications, ensuring efficient data processing and retrieval.

1.2: As a front-end developer, I want to implement the interface for the machine learning module to provide users with an interactive and user-friendly experience.

1.3: As a backend developer, I want to expand the API suite to support the enhanced database functionalities, allowing seamless interaction and integration across different system components.

1.4: As a front-end developer, I want to integrate the new APIs into the front-end system to ensure all components communicate effectively and data flows seamlessly.

1.5: As a software engineer, I want to evaluate various machine learning algorithms to determine which one performs optimally for our specific needs.

1.6: As a software engineer, I want to integrate the selected machine learning algorithm into the production codebase to enable practical application and real-world testing.

1.7: As a software engineer, I want to assemble all components into a cohesive system to ensure functionality and smooth operation across all parts of the application.

1.8: As a developer, I want to refactor code for efficiency and maintainability to ensure the application is easy to manage and update.

1.9: As a software engineer, I want to complete and refine the project documentation to provide clear and comprehensive guidance for users and future developers.



1.10: As a software engineer, I want to conduct comprehensive backend testing to ensure that all server-side functions operate correctly and meet performance standards.

1.11: As a software engineer, I want to conduct rigorous front-end testing to ensure the user interface is reliable, efficient, and user-friendly.

## Project Test Plan

Test at the end of the project, Week 12

## Milestones and Deadline

### Schedule:

Week 1&2: 1.0

Week 3: 1.1

Week 4: 1.2

Week 5: 1.3

Week 6: 1.4

Week 7&8: 1.5

Week 9: 1.6

Week 10: 1.7

Week 11: 1.8 & 1.9

Week 12: 1.10 & 1.11

Deadline? Presentation?

## Risk Analysis

1. Full Time Summer Job
2. Vacation (no date yet)

## Project Progress



# Feature Implementation

## Design

<https://github.com/lohnerj/ase-392ResearchProject/tree/main/Design>   
(<https://github.com/lohnerj/ase-392ResearchProject/tree/main/Design>)

## Code

<https://github.com/lohnerj/ase-392ResearchProject/tree/main/marketapp/lib>   
(<https://github.com/lohnerj/ase-392ResearchProject/tree/main/marketapp/lib>)

## Tests

<https://github.com/lohnerj/ase-392ResearchProject/tree/main/marketapp/test>   
(<https://github.com/lohnerj/ase-392ResearchProject/tree/main/marketapp/test>)

## Document

<https://github.com/lohnerj/ase-392ResearchProject/tree/main/Documentation>   
(<https://github.com/lohnerj/ase-392ResearchProject/tree/main/Documentation>)

## Week 1

### Summary

Made Basic Front/Backend design for the project:

- Text Documentation
- Canva Link to flowchart

### Milestones or risks in this week

Milestone 1.0 is not due until next week, but it is basically finished.

## Week 2

### Summary

Revisited Documentation for architecture

Finished 1.1

Started on Milestone 1.1

### Milestones or risks in this week

Milestone 1 Completed



## Week 3

### Summary

Wrote code for the new SQL database

### Milestones or risks in this week

Milestone 1.1 is 90% complete, will finish this weekend

## Week 4

### Summary

Writing the front-end code for this ML part of this application, not totally finished, because I have to leave some space open for the ML part, which hasn't been started yet.

### Milestones or risks in this week

Milestone 1.2

Risks: Starting an internship

## Week 5

### Summary

Started work on the api suite with database, fell a little behind. Will finish saturday/sunday.

### Milestones or risks in this week

1.3: As a backend developer, I want to expand the API suite to support the enhanced database functionalities, allowing seamless interaction and integration across different system components.

Risk: Started Internship

## Week 6

### Summary

Worked on using the new apis to work with the front-end buttons.

### Milestones or risks in this week

1.4: As a front-end developer, I want to integrate the new APIs into the front-end system to ensure all components communicate effectively and data flows seamlessly.

## Week 7&8

### Summary



Tested out a couple of algorithms, deciding that simple is best and that I need speed over accuracy mainly, but obviously the accuracy cannot be too low. Settling with a simple linear regression for now.

### Milestones or risks in this week

1.5: As a software engineer, I want to evaluate various machine learning algorithms to determine which one performs optimally for our specific needs.

## Week 9

### Summary

Worked on implementing the ML algorithm into the code, been having some issues. Will take some time to out of the next week to get it fully working,

### Milestones or risks in this week

1.6: As a software engineer, I want to integrate the selected machine learning algorithm into the production codebase to enable practical application and real-world testing.

## Week 10

### Summary

Finished up 1.6, and then made sure all components are working together as intended. Basically integration testing.

### Milestones or risks in this week

1.7: As a software engineer, I want to assemble all components into a cohesive system to ensure functionality and smooth operation across all parts of the application.

## Week 11

### Summary

Finished refactoring and integrating the code. Started creating the proper documentation.

### Milestones or risks in this week

1.8: As a developer, I want to refactor code for efficiency and maintainability to ensure the application is easy to manage and update.

1.9: As a software engineer, I want to complete and refine the project documentation to provide clear and comprehensive guidance for users and future developers.

